



TOBI SEGUN OLUWATEGBE

AI Engineer | M.Sc. Data Analytics Candidate

<https://www.linkedin.com/in/tobi-segun-o-01893b1a3/>
+49 157 33698894 tegbetobi2@gmail.com Lower Saxony, Germany
Website : <https://tegbetobi.github.io/portfolio/>

SKILLS

Programming & Backend Development

- Python, SQL, Linux, Git
- FastAPI, Flask, REST APIs, API Integration

Generative AI & Agent Development

- AI Agents, Agentic Workflows
- LangChain, LangGraph, HuggingFace
- OpenAI, Gemini APIs, Large Language Models
- Tool / Function Calling, Structured Outputs
- Agent State Management, Memory Architectures, Long-Term Memory
- Prompt Engineering, Prompt Optimization
- LLM Evaluation, Agent Evaluation, Prototype Testing
- Retrieval-Augmented Generation (RAG), Knowledge Retrieval, AI Automation
- Vector Databases, Embeddings, Semantic Search

AI Research & Experimentation

- AI Tool & Framework Evaluation
- Model / Agent Benchmarking
- Experimental Comparison of AI Approaches
- Retrieval Quality Evaluation
- Proof-of-Concept Development

Cloud & DevOps

- Azure, AWS, Docker
- CI/CD Pipelines (GitLab CI/CD, Jenkins, GitHub Actions)

LANGUAGES

- English (C2 – Native)
- German (B2 – Advanced)

EDUCATION

Master of Science (M.Sc.) in Data Analytics
Stiftung Universität Hildesheim, Germany |
04/2024 – Present

**Bachelor of Engineering (B.Eng.)
Electrical & Electronics Engineering**
Covenant University, Nigeria | 2016 – 2021

WORK AUTHORIZATION

Eligible to work in Germany as a Working Student and Intern.

PROFESSIONAL EXPERIENCE

Azure Technical Support Engineer

Microsoft (via Tek Knowledge - Microsoft Delivery Partner)

03/2023 - 04/2024

- Improved Azure cloud performance by 30% by monitoring VM resources, troubleshooting compute issues and optimizing infrastructure utilization.
- Reduced cloud infrastructure costs by 25% through resource utilization analysis, right-sizing strategies.
- Resolved 80% of enterprise support incidents within two days by diagnosing cloud environments and meeting strict SLA requirements as well as generating root-cause analysis across enterprise cloud environments.

Cloud Engineer

Lotus Beta Analytics, Nigeria

01/2022 - 03/2023

- Migrated on-premises applications to Microsoft Azure using cloud-native services, improving scalability and deployment flexibility.
- Built and maintained CI/CD pipelines with Azure DevOps, increasing deployment efficiency and consistency across environments.
- Implemented secure networking solutions, including VPN and SSL configurations, to meet client security and compliance requirements.

PROJECTS

Sports Travel Agent - https://github.com/tegbetobi/Sports_Travel_AI_Agent

- Built an AI agent system for automated sports travel planning using Python, Pydantic, FastAPI, LangChain, and REST API integrations.
- Designed stateful agent workflows with tool calling, structured outputs, prompt engineering, and conditional routing.
- Evaluated agent behavior and tool-calling reliability to identify failure cases and improve response quality.
- Deployed the application with Docker, AWS ECS Fargate, and Jenkins CI/CD.

Medical RAG Chatbot - https://github.com/tegbetobi/RAG_Chatbot

- Built a RAG-based medical chatbot using FastAPI, HuggingFace embeddings, and vector search.
- Implemented document ingestion, chunking, embeddings, semantic search, and top-k retrieval.
- Evaluated retrieval using Recall@K, MRR, retrieval latency and answer relevance and compared chunking and prompt configurations to reduce hallucinations and improve response quality.
- Compared chunking strategies improving Recall@5 by roughly 15%, reducing average retrieval latency

CERTIFICATIONS & RELEVANT COURSEWORK

- **LLM Engineering, RAG & AI Agents Masterclass (Udemy, 2025)** – Hands-on training covering LLM application development, RAG pipelines, vector databases, prompt engineering, tool calling, and AI agent orchestration.
- **University Coursework** : Advanced Machine Learning, Large Language Models